

Biology, 2013 EC, natural science, entrance exam, round 1

Grade 12 · Biology · 2013 E.C.

Time allowed: 120 minutes

Questions: 87

Total marks: 87

Name _____ Grade _____ ID No. _____ Date _____

1. Which one of the following is true about the importance of Lucy in resolving debates about human evolution? (1)
 - A. Big brains came before bipedalism.
 - B. Lucy had a brain size of 1.8% of her body mass.
 - C. Bipedalism came before big brains.
 - D. Lucy was partially an arboreal primate
2. The quick withdrawal of your hand from a hot plate is an example of (1)
 - A. fixed action pattern
 - B. innate behavior
 - C. biological clock
 - D. key stimulus
3. Which one of the following processes describes a biological clock? (1)
 - A. Removal of one's thumb from hot objects quickly
 - B. The growing of plants towards light
 - C. Circannual migration of birds from temperate to tropical areas
 - D. Kinesis and taxes of woodlice and other simple animals
4. Identify learned behavior from the following. (1)
 - A. Dog trained to hunt
 - B. Honey bee making honey
 - C. Child withdraws hand from hot things
 - D. Ants making nest
5. Which one of the following features is unique to bacteria? (1)
 - A. Presence of permeable membrane
 - B. Being unicellular
 - C. Absence of nuclear membrane
 - D. Having flagella for locomotion
6. It is possible to produce human insulin in a bacterial strain. The insulin gene should be inserted in a plasmid and the recombinant DNA plasmid should be inserted into a bacterium. What is the role of the bacterium in this process? (1)
 - A. The source of insulin and modification
 - B. The source of plasmid and insulin production
 - C. Multiplying the recombinant DNA molecule
 - D. Joining the gene of insulin with the gene of plasmid
7. Bacteria can cause diseases in all of the following mechanisms EXCEPT (1)
 - A. Changing the genetic structure of host cells
 - B. Entering living cells and disrupting metabolic systems
 - C. Releasing toxins as they multiply
 - D. Invading and growing on the tissues of organs

-
8. Functional diseases differ from infectious diseases in that they (1)
- | | |
|--|---|
| A. Are caused by microorganisms | B. Occur in the brain and heart |
| C. Are more common in females than males | D. Show no signs of damage on the organ |
-
9. The social impact of AIDS can be mainly revealed by (1)
- | | |
|---------------------------|------------------------|
| A. Fear of stigmatization | B. Loss of appetite |
| C. Reduced labor force | D. Loss of body weight |
-
10. Which one of the following is a DNA virus? (1)
- | | |
|-------------------|---------|
| A. Herpes complex | B. HIV |
| C. Hepatitis C | D. H1N1 |
-
11. Which one of the following is a correct comparison of lytic and lysogenic cycles of virus multiplication? (1)
- A. Lysogenic cycle releases new viruses using only chronic cycle
- B. Lytic cycle involves incorporation of viral DNA into the host DNA
- C. Lytic cycle releases new viruses only by splitting the host cell
- D. Both cycles damage the host cell DNA following infection
-
12. In an HIV/AIDS community screening, students were asked to take part in voluntary testing. The following discussion presents the views of four grade 12 students named A, B, C, and D towards HIV testing. Student A: I should not take the testing; it is simply boring. Student B: We should take the testing since we are educated; we should be exemplary. Student C: If we get tested and get positive, what should we do? So, let us not get tested. Student D: You know, this HIV/AIDS testing is simply a business. We should not be cheated. Which student do you think is correct? (1)
- | | |
|--------------|--------------|
| A. Student B | B. Student A |
| C. Student D | D. Student C |
-
13. Which event of cellular respiration takes place in the cytoplasm of eukaryotic cells? (1)
- | | |
|---|---|
| A. Production of pyruvate from glucose | B. Reaction of acetyl CoA with oxaloacetate |
| C. Production of ATP through chemiosmosis | D. Reaction of pyruvate with coenzyme A |
-
14. Suppose two moles of glucose is burned aerobically. How many moles of ATP can be produced by substrate level phosphorylation in Krebs cycle? (1)
- | | |
|-------|------|
| A. 4 | B. 6 |
| C. 10 | D. 8 |
-
15. During alcoholic fermentation, CO₂ is released when (1)
- | | |
|-------------------------------|-------------------------|
| A. Acetaldehyde is reduced | B. Pyruvate is oxidized |
| C. Pyruvate is decarboxylated | D. Ethanol is oxidized |
-
16. During anaerobic respiration, reduced NAD is required to (1)
- | | |
|---|----------------------------------|
| A. Split pyruvate to CO ₂ and acetaldehyde | B. Reduce pyruvate to lactate |
| C. Convert lactate into pyruvate | D. Reduce pyruvate to acetyl CoA |
-
17. Why do we get tired and fatigued during heavy exercise such as long distance running? It is due to (1)
- | | |
|-------------------------------------|--------------------------------|
| A. Accumulation of CO ₂ | B. Accumulation of lactic acid |
| C. High rate of aerobic respiration | D. Shortage of glucose |

-
- 18.** If one base in the six triplets of the gene coding for one of the four polypeptides in the haemoglobin molecule alters the triplet from GAG to GTG, which one of the following biological terms best describes the above phenomenon? (1)
- A. Mutation
B. Cloning
C. Translation
D. Transcription
-
- 19.** Which one of the following is the basic characteristic of mutation? It is (1)
- A. Spontaneous
B. Planned
C. Regular
D. Expected
-
- 20.** The theory that suggests' life can evolve autonomously from non-living objects' is known as (1)
- A. Cosmozoan theory
B. Spontaneous generation theory
C. Biochemical theory
D. Eternity theory
-
- 21.** Which one of the following is true about neo-Darwinian theory? (1)
- A. There is natural variation in features and variations which are heritable
B. All species tend to produce more offspring than can possibly survive
C. It takes into account genetics and animal behavior to understand evolution
D. Development of new feature is important for survival
-
- 22.** Which one of the following statements describes the evolutionary theory of Lamarck? (1)
- A. Acquired characteristics cannot be inherited
B. Structures commonly used become more developed
C. Origin of species happens only through natural selection
D. Species started complex and gradually become simple
-
- 23.** The behavior which trainers look, if modified, could lead to the desired behavior and reward such a behavior is (1)
- A. Innate
B. Reflex
C. Instinct
D. Operant
-
- 24.** Which one of the following examples of behavior describes when an animal stops responding to stimulus or cue after a period of repeated exposure? (1)
- A. Imprinting
B. Habituation
C. Reflex
D. Conditioning
-
- 25.** A police man trained a dog on how to find explosive materials through sniffing, on another instant the same police man observed a week old puppy following him on the roadside. Which one of the following is correct about the behaviors of the animals in the observation? (1)
- A. The first case is genetically encoded and the second is adaptive
B. Both of the cases indicate natural behavior of the animals
C. Both of the cases indicate learned behavior of the animals
D. The first case develops through trial and the second is genetically coded
-
- 26.** Which one of the following behaviors is a courtship behavior? (1)
- A. Touch
B. Ritual fighting
C. Scent marking
D. Defending a den

27. The Ethiopian wolf (*Canis simensis*) lives and hunts in packs. During patrolling their habitat, both male and female wolves of the group maintain their area with urine containing pheromone. This pattern of behavior best exemplifies

- A.** Courtship behavior **B.** Territorial behavior
- C.** Operant behavior **D.** Learned behavior

28. The terrestrial biome characterized by cold temperature, dry precipitation, frozen soil, and inhabited by migrating birds is _____ (1)

- A. Boreal forests
B. Deserts
C. Tropical montane forests
D. Tundra

29. Which one of the following explains biodiversity? (1)

- A. The variability among living organisms, within species, and of ecosystems
- B. The large number of human races in the world
- C. The similarity and difference of organisms which live in different ecosystems
- D. Large number of organisms in the same ecosystem

30. What is the major difference that viruses have with free-living cells? Viruses (1)

- A. Reproduce only asexually
B. Lack genetic material of their own
C. Do not function outside of a living cell
D. Lack locomotory structure

31. Why is nutrient recycling important in an environment? Because (1)

- A. The amount of nutrients in the environment is limited
- B. It adds new types of molecules into the environment
- C. Nutrients in an environment will be depleted if not recycled
- D. Nutrients should be exchanged between the Earth and other planets

32. Which one of the following orders of life processes correctly illustrates the occurrence of the nutrient cycle? (1)

- A.** Feeding → Death and decomposition → Excretion → Respiration and breathing
B. Feeding → Excretion → Respiration and breathing → Death and decomposition
C. Death and decomposition → Feeding → Excretion → Respiration and breathing
D. Respiration and breathing → Feeding → Excretion → Death and decomposition

33. Which of the following processes in the phosphate cycle DOES NOT return phosphate ions to the soil? (1)

- A.** Weathering of rocks **B.** Use of fertilizers
C. Decomposition of dead animals **D.** Cellular respiration

34. A previously bare land was gradually occupied by lichen and mosses, annual herbs, perennial herbs, shrubs, and finally by forests. Which one of the following best describes the process indicated above?

- A. Afforestation
B. Succession
C. Climax community
D. Natural selection

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- 35.** This item is based on the table below which shows population growth trend. Which one of the following is correct about X and Y? (1)
- A. X is exponential growth because the numbers increase by the same fixed amount in each time period and Y is arithmetic growth because the population doubles in each time period
 - B. X and Y cannot be compared because each of them grows differently with incomparable proportions
 - C. X is arithmetic growth because the numbers increase by the same fixed amount in each time period and Y is exponential growth because the population doubles in each time period
 - D. Both X and Y are exponential growth since each of them increases regularly without interruption
-
- 36.** *Drosophila melanogaster* are convenient experimental animals in genetics because they (1)
- A. Have 23 pairs of chromosomes per cell
 - B. Are small animals with a short life cycle
 - C. Are too cheap to breed in large numbers
 - D. Have chromosomes that can't be easily detected
-
- 37.** Suppose a heterozygous brown-eyed father is married to a blue-eyed mother. What would be the proportion of homozygous children for eye color? (Hint: Brown is dominant over blue) (1)
- A. 50%
 - B. 100%
 - C. 75%
 - D. 25%
-
- 38.** If organisms share very similar molecules such as DNA, then they must be closely related evolutionarily. Which evidence of evolution supports the above information? (1)
- A. Comparative embryology
 - B. Comparative anatomy
 - C. Comparative physiology
 - D. Comparative biochemistry
-
- 39.** Which one of the following CANNOT be used as evidence of evolution through paleontology? (1)
- A. Stomach
 - B. Teeth
 - C. Hair
 - D. Bones
-
- 40.** The process in which individuals with a particular heritable trait survive better than others is (1)
- A. Adaptation
 - B. Selective breeding
 - C. Evolution
 - D. Natural selection
-
- 41.** Which type of selection favors individuals at both extremes of a range to have some advantage over those displaying the mean value? (1)
- A. Stabilizing selection
 - B. Directional selection
 - C. Natural selection
 - D. Disruptive selection
-
- 42.** Which one of the following is true about human evolution? (1)
- A. Humans and chimpanzees both evolved from a common ancestor
 - B. Humans evolved from chimpanzees
 - C. Humans evolved from a different line of evolution than great apes
 - D. Humans evolved from great apes
-
- 43.** Which one of the following organic molecules is used to make hemoglobin? (1)
- A. Glycogen
 - B. Proteins
 - C. Lipids
 - D. Nucleic acids

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- 52.** Suppose there are two communities in a village. Community A actively participates in HIV/AIDS awareness creation, HIV prevention, and gets blood tests for HIV, whereas community B frequently requests food aid, participates only in afforestation activities, and HIV-positive individuals keep the secret to themselves. In which community do you think that the incidence of HIV/AIDS is most probably high? (1)
- A. In both communities
B. In community B
C. Impossible to decide
D. In community A
-
- 53.** Which organic molecules are formed when two fatty acid molecules are bound to glycerol, and the place of the third is taken by a phosphate group? (1)
- A. Waxes
B. Glycolipids
C. Triglycerides
D. Phospholipids
-
- 54.** All are the functions of lipids EXCEPT (1)
- A. Thermal insulation
B. Waterproofing
C. Respiratory substrates
D. Determining a particular feature
-
- 55.** Three different cells are in the process of transporting materials across the cell membrane as indicated below: (1)
Cell 1 = water from high concentration to low concentration
Cell 2 = Na⁺ from high concentration to low concentration
Cell 3 = glucose from low concentration to high concentration
Which types of transport mechanisms do the above cells use, respectively?
- A. Osmosis, facilitated diffusion, active transport
B. Osmosis, active transport, facilitated diffusion
C. Facilitated diffusion, osmosis, active transport
D. Facilitated diffusion, active transport, osmosis
-
- 56.** Suppose you took a container and divided it into two halves by a membrane which is permeable to water and labeled the two halves as A and B. Finally, you added hypertonic solution in A and hypotonic solution in B. What will be the expected result of this experiment? (1)
- A. Rise of the water level in both halves
B. Rise of the water level in B
C. Rise of the water level in A
D. No change in both halves
-
- 57.** According to the fluid-mosaic model of the plasma membrane (1)
- A. The membrane has uniform composition of proteins
B. Glycoproteins form a double layer
C. Phospholipid bilayer forms the basic structure of the membrane
D. Peripheral proteins are tightly attached to the membrane
-
- 58.** A class of grade 11 students were given two solutions A and B of unknown concentration separated by a semi-permeable membrane. The teacher instructed them to observe any changes in the water levels of the two solutions. After a few minutes, the students observed that the water level in solution A had decreased. What could the students conclude from their observation? (1)
- A. Solution B had a positive water potential
B. Solution A had a higher water potential
C. The two solutions were isotonic
D. Solution A was hypertonic

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- 59.** A student did an experiment and found out that catabolism of amino acid produces toxic ammonia in the liver cells, which gradually gets converted into urea and excreted through urine. What would be the result of the experiment if osmosis was blocked? The liver cells (1)
- A. Gain water and become turgid
 - B. Excrete more ammonia and urea
 - C. Become overactive and produce more energy
 - D. Become damaged and lose functionality
-
- 60.** Consider the following key events in the study of cell biology: I. Hooke describes 'cells' in a cork. II. Leeuwenhoek discovers protozoa. III. Schleiden and Schwann propose a cell theory. IV. Jansen builds the first compound microscope. Which one of the following represents the correct order from ancient to the recent? (1)
- A. I, II, III, and IV
 - B. IV, I, II, and III
 - C. III, II, I, and IV
 - D. II, III, IV, and I
-
- 61.** One of the differences of prokaryotic and eukaryotic cells is their cell wall. This is because the cell wall of prokaryotic cell is (1)
- A. The site of energy production
 - B. Impermeable
 - C. Stronger than eukaryotic cells
 - D. Made from sugar and amino acid
-
- 62.** Consider the following features of cells: I. Have pili II. Have a chromosome III. Have circular DNA IV. Have larger ribosomes Which of the above features of cells can be included in eukaryotic cells? (1)
- A. II and III
 - B. I and II
 - C. II and IV
 - D. I and III
-
- 63.** Which of the following is correct about the structure of the cell membrane? (1)
- A. Phospholipid molecules form a transport channel
 - B. Protein molecules form a double layer
 - C. The membrane has a fluid nature that is regulated by cholesterol
 - D. Proteins form the outer layer while phospholipids form the inner layer
-
- 64.** Which of the following is a correct match of cellular organelles and their corresponding function? (1)
- A. Vacuole - strength and elasticity
 - B. Mitochondria - ATP synthesis
 - C. Lysosome - modification of proteins
 - D. Golgi body - digesting foreign cells
-
- 65.** What structural arrangements make chlorophyll suitable for its function? (1)
- A. The floating of the photosystems and the electron transport chain in the cytosol
 - B. The fixing of the photosystems and the electron transport chain on the thylakoid membrane
 - C. The structural differences between Photosystem I and Photosystem II
 - D. The long distance between Photosystem I and Photosystem II
-
- 66.** Why does photorespiration take place in a hot environment? Because plants close their stomata and face (1)
- A. H₂O shortage
 - B. O₂ accumulation
 - C. CO₂ shortage
 - D. O₂ shortage

-
- 67.** Which of the following is correct about the process of glucose production in the light-independent reactions of photosynthesis? (1)
- A. A total of 9 ATP molecules are synthesized.
 - B. CO₂ reacts with GP to form RuBP.
 - C. One TP is directly converted to one glucose molecule.
 - D. GP is reduced to TP by reduced NADP.
-
- 68.** Which of the following is a correct description of the CAM pathway of photosynthesis? (1)
- A. CO₂ is stored in the form of malate during the night.
 - B. The Calvin cycle takes place during the night.
 - C. The Calvin cycle takes place in the bundle sheath cells during the day.
 - D. CO₂ is stored in the bundle sheath cells during the night.
-
- 69.** Which one of the following statements explains how an enzyme lowers the activation energy of a reaction? By (1)
- A. Enabling the reaction to occur at lower physiological temperature
 - B. Making the individual substrate molecules more reactive
 - C. Lowering the kinetic energy of the reacting molecules
 - D. Changing its tertiary structure into secondary protein structure
-
- 70.** Which one of the following enzymes is used as washing powders? (1)
- | | |
|----------------|--------------|
| A. Biochymosin | B. Lipase |
| C. Cellulase | D. Pectinase |
-
- 71.** What happens when the substrate molecule binds to the active site of an enzyme? (1)
- A. The substrate and the enzyme will bond permanently.
 - B. They form a stable intermediate substance.
 - C. The amino acids in the active site become unreactive.
 - D. The chemical bonds of the substrate are distorted.
-
- 72.** Which of the following vitamins is used as a component of cofactor? (1)
- | | |
|------------------|---------------|
| A. Thiamin | B. Riboflavin |
| C. Ascorbic acid | D. Retinol |
-
- 73.** How does crossing over during meiosis bring variation in species? (1)
- A. Increasing the amount of genes in one of the parents during meiosis
 - B. Combination of genes in the gametes that are not found in either parent
 - C. Combining sex and somatic cell chromosomes
 - D. Creating genes that are not found in both parents due to meiosis
-
- 74.** How does our body synthesize some of the amino acids in our body? (1)
- A. From fatty acids we consume
 - B. From intermediate molecules of different pathways
 - C. From the breakdown of pre-existing proteins
 - D. From the biological breakdown of our own cells

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- 75.** This item is based on the following list of events during cloning of an insulin gene into plasmid. I. Obtaining insulin-producing cells and plasmid from organisms II. Cutting the insulin gene and plasmid III. X IV. Inserting the recombinant plasmid into bacteria V. Production of insulin by bacteria Which one of the following best represents the step listed as 'X'?
- (1)
- A. Inserting the insulin gene into a bacterium
 - B. Inserting the open plasmid into a bacterium
 - C. Fusing the insulin and plasmid genes to the cell wall of bacteria
 - D. Combining the insulin gene into the opened plasmid
-
- 76.** Consider the DNA molecule containing: Strand 1: T A C G T A Strand 2: A T G C A T What will be the sequence of the new DNA if the DNA replicates semi-conservatively?
- (1)
- A. ATGCAT UACGUA
 - B. TACGTA ATGCAU
 - C. ATGCAT ATGCAT
 - D. TACGTA ATGCAT
-
- 77.** During translation, the
- (1)
- A. tRNA serves as a template of the DNA molecule.
 - B. ribosome moves forward over two codons at a time.
 - C. mRNA brings amino acids into the ribosomes.
 - D. tRNA with complementary anticodon binds on the codon of mRNA.
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- 78.** The step of the scientific method which comes after analyzing results and drawing a conclusion is
- (1)
- A. Doing background research.
 - B. Constructing a hypothesis.
 - C. Designing and carrying out an experiment.
 - D. Accepting or rejecting the hypothesis.
-
- 79.** In a group of individuals in which Group A was a control and Group B was an experimental group, a student gave a placebo to the control group and an unknown drug to the experimental group. At the end, the experimental group got better. Which one of the following is true about the result?
- (1)
- A. The drug has an effect.
 - B. Placebo has a better effect.
 - C. The experiment was wrong.
 - D. Conclusion cannot be reached.
-
- 80.** A biologist conducted a simple experiment to investigate the effect of pH on salivary amylase. She added 5cm³ amylase solution into 5cm³ starch suspension with a neutral pH in a test tube. After ten minutes, she tested it with iodine reagent and the solution turned into blue-black color. What is the dependent variable of this experiment?
- (1)
- A. Applying iodine treatment
 - B. pH of the solution
 - C. Adding salivary amylase
 - D. Formation of blue-black color
-
- 81.** Which one of the following pairs of biological tools and their functions is FALSE?
- (1)
- A. Petri dish - used for culturing microorganisms
 - B. Balances - used for measuring mass
 - C. Quadrant - to separate solids from liquids
 - D. Optical microscope - use beam of light to produce magnified image

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- 82.** Grade 12 students went to a field and collected plant specimens for identification in the laboratory. They wanted to preserve the plant parts for some time and specimens can be analyzed later. Which one of the following biological tools is used for their activities? (1)
- A. Plant press
B. Pitfall traps
C. Theodolite
D. Quadrat
-
- 83.** Some amino acids are synthesized using the transamination process in the cell. This activity more describes the cell's (1)
- A. Replication
B. Energy production
C. Catabolism
D. Anabolism
-
- 84.** Identify the statement that correctly explains the factors of an enzyme activity. (1)
- A. Excessive heat breaks the ionic bonds that hold the tertiary structure of the enzyme protein.
B. Hydrogen ion concentration affects the ionic bonds of an enzyme.
C. Increasing concentration of substrates does not affect enzyme activity.
D. Inhibitors increase enzyme activity by increasing their turnover rate.
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- 85.** Which one of the following is true about allosteric regulation? (1)
- A. Enzymes are activated when activators attach to the active site.
B. Inhibition by competitive inhibitors occurs when they interact at the allosteric site.
C. Activation by activators occurs when they interact at the allosteric site.
D. Allosteric inhibition occurs when inhibitors attach to the active site.
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- 86.** If an enzyme from thermophilic bacteria, which catalyzes glucose 6-phosphate into fructose 6-phosphate in the process of glycolysis, is subjected to below the optimum temperature, the rate of the reaction will decrease. This is because (1)
- A. The particles have more kinetic energy.
B. The enzyme is denatured.
C. More collisions between enzyme and glucose 6-phosphate occurred.
D. Less enzyme-glucose 6-phosphate complexes are formed.
-
- 87.** Cyclooxygenase-2 is an enzyme that catalyzes the conversion of arachidonic acid to prostaglandins, which promote pain and fever in our body. The painkiller ibuprofen acts as a non-competitive inhibitor of cyclooxygenase-2, reducing the production of prostaglandins. What would happen if the concentration of ibuprofen increased? (1)
- A. Pain of the body is decreased.
B. Production of prostaglandin is increased.
C. Concentration of arachidonic acid is decreased.
D. Substrate affinity of cyclooxygenase-2 is higher.

— END OF EXAMINATION —